

Ontology-Constrained Vector Retrieval

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TABLE I

DATASET AND BENCHMARK CHARACTERISTICS. INDEXED TYPES ARE DISTINCT DIRECTLY ASSERTED ONTOLOGY TYPES REPRESENTED AMONG INDEXED ENTITIES. CONSTRAINT FAMILIES ARE DISTINCT SETS OF ACCEPTABLE TARGET CLASSES USED BY THE BENCHMARK QUERIES.

Dataset	Entities	Types	Constraints	Queries
ATMONTTO	36,655	7	5	50
Brick	19,388	85	31	50
DBpedia	23,189	69	16	50

Abstract—

Index Terms—ontology, vector database, semantic retrieval, filtered search, knowledge graph

I. INTRODUCTION

II. RELATED WORK

III. PROBLEM FORMULATION

IV. ONTOLOGY-CONSTRAINED RETRIEVAL SYSTEM

V. EXPERIMENTAL SETUP

A. Datasets and Benchmarks

We evaluate ontology-constrained vector retrieval across three ontology-backed datasets spanning substantially different domains: NASA’s Air Traffic Management ontology (ATMONTTO), the Brick building ontology with selected Mortar building models, and a DBpedia-derived benchmark of U.S. civic places and institutions. Table I summarizes the indexed datasets and benchmark composition. Each dataset is evaluated using 50 natural-language queries, yielding 150 queries in total. Each query is paired with one or more ontology classes that define the acceptable result type.

ATMONTTO. Our primary aviation-domain dataset uses NASA’s Air Traffic Management ontology and associated National Airspace System data [?]. The retained source consists of 29 Turtle files containing 2,931,368 unique RDF triples after graph de-duplication. From this graph, we index 36,655 entities spanning six represented operational families:

airspace route segments, international airports, intersection fixes, latitude–longitude fixes, METAR weather reports, and TAF meteorological conditions. The benchmark contains 50 queries grouped into five constraint families covering route segments, navigation fixes, airports, METAR reports, and TAF forecast conditions.

Brick. The building-domain dataset combines the Brick ontology TBox [?] with 45 selected Mortar building/model graphs. The combined graph contains 121,209 unique RDF triples and yields 19,388 indexed entities. These entities include building equipment, locations, and measurement/control points and collectively represent 85 directly asserted ontology types. The 50-query benchmark uses 31 distinct constraint families, including equipment classes such as chillers and air-handling units, locations such as rooms and HVAC zones, and measurement or control-point classes such as supply-air temperature sensors and setpoints.

DBpedia U.S. Civic Geography. To evaluate the approach in a broader and less domain-specific setting, we construct a DBpedia-based benchmark from U.S. civic places and institutions [?]. The retained dataset contains 23,189 indexed entities retrieved from the public DBpedia SPARQL endpoint, including cities, towns, villages, schools, universities, hospitals, libraries, religious buildings, dams, stadiums, and other civic structures or venues. Unlike ATMONTTO and Brick, this benchmark is a constructed entity subset rather than a retained RDF graph snapshot. The final benchmark contains 50 queries across 16 constraint families.

B. Entity Representation and Vector Index

For each dataset, we extract typed entities and retain their direct ontology types together with the transitive superclass closure induced by `rdfs:subClassOf`. For an entity e , we denote its direct types by T_e and its type closure by T_e^+ . The closure allows a query constraint to match both an explicitly asserted class and an appropriate superclass represented in the ontology hierarchy.

Each entity is converted into a textual *entity card* combining its label, ontology-type information, and selected dataset-specific attributes. For ATMONTO, these attributes include information such as airport names and identifiers, navigation-fix identifiers, route information, coordinates, and selected meteorological fields. Brick and DBpedia entity cards use analogous attributes available within their respective source data.

Entity cards are embedded and stored in Qdrant 1.17.0 using cosine similarity. Each indexed point contains the embedding vector together with the entity identifier and two ontology-derived payload fields: `types`, containing directly asserted classes, and `types_closure`, containing the direct classes and their represented superclasses. Both fields are stored using Qdrant keyword payload indexes. The final collections are built with HNSW vector indexes and are verified as fully indexed before benchmark evaluation.

C. Embedding Models

We evaluate two pretrained sentence-transformer embedding models: `all-MiniLM-L6-v2` and `bge-large-en-v1.5`. MiniLM produces 384-dimensional embeddings, while BGE produces 1024-dimensional embeddings. Entity cards and queries are embedded independently for each model, producing separate Qdrant collections under otherwise identical retrieval conditions. Embeddings are normalized prior to indexing and cosine similarity is used for nearest-neighbor ranking.

D. Retrieval Conditions

For each query, we evaluate three retrieval strategies with $k = 5$.

Unconstrained retrieval performs standard vector search over the collection without supplying an ontology filter and returns the five highest-ranked results.

Pre-hoc ontology-constrained retrieval supplies the query’s ontology constraint directly to Qdrant as part of the vector retrieval request. The filter requires the `types_closure` payload to match at least one acceptable target class. Thus, only entities satisfying the requested ontology constraint are eligible to appear in the returned result set, while vector similarity determines their ranking.

Post-hoc ontology filtering first performs unrestricted vector retrieval and subsequently checks the returned entities against the same ontology constraint outside the database. Candidate limits of 25, 100, 200, 400, and 2,000 are evaluated. Retrieval begins with an unrestricted top-25 request and increases the requested prefix in increments of 25 until five valid entities are obtained or the configured candidate limit is reached. Only newly encountered candidates are evaluated after each expansion.

This comparison isolates the difference between supplying the ontology-derived constraint during database retrieval and attempting to recover type-valid results from an unrestricted similarity ranking afterward.

E. Evaluation Metrics

We evaluate ontology-type correctness using two complementary metrics. For a query q , let $R_k(q)$ denote the first k returned entities and let C_q denote the query’s acceptable ontology classes. *Valid@ k* is defined as

$$\text{Valid@}k(q) = \frac{1}{k} \sum_{e \in R_k(q)} \mathbb{I}[T_e^+ \cap C_q \neq \emptyset], \quad (1)$$

where missing result positions count as invalid. *Valid@ k* therefore measures the fraction of returned positions that satisfy the specified ontology-type constraint.

We additionally report *AllValid@ k* , which indicates whether the complete returned result set is type-valid:

$$\text{AllValid@}k(q) = \mathbb{I}[\text{Valid@}k(q) = 1]. \quad (2)$$

Reported *AllValid@ k* values are averaged across queries and therefore represent the fraction of benchmark queries for which all k returned entities satisfy the ontology constraint.

We also report the mean vector-similarity score of the returned results and client-observed retrieval latency. Query latency measures the Qdrant retrieval request and response, including local HTTP communication, database query processing, payload retrieval, and response deserialization. Query embedding time is excluded. For post-hoc retrieval, latency is the sum of the Qdrant requests required as the candidate prefix is expanded; local ontology-validity checking between requests is not included. Unless otherwise stated, all reported validity metrics use $k = 5$. ““

VI. RESULTS

A. Quantitative Retrieval Performance

We evaluate retrieval using two complementary measures of ontology-type correctness. *Valid@5* is the fraction of the five returned entities that satisfy the specified ontology-derived type constraint. *AllValid@5* is the fraction of queries for which all five returned entities satisfy the constraint. We additionally report the mean vector-similarity score of the returned results and client-observed retrieval latency.

Table II compares unconstrained vector retrieval, pre-hoc ontology-constrained retrieval, and post-hoc filtering at a maximum candidate limit of 2,000. Pre-hoc retrieval achieved *Valid@5* and *AllValid@5* of 1.000 across every dataset and both embedding models. In contrast, unconstrained retrieval frequently returned entities that were semantically related to the query but did not satisfy the requested ontology type.

Averaged across the three datasets, unconstrained *Valid@5* was 0.6387 with `all-MiniLM-L6-v2` and 0.7093 with `bge-large-en-v1.5`. Corresponding *AllValid@5* values were 0.5133 and 0.5800, indicating that only about half of unconstrained queries produced a completely type-valid set of five results. Pre-hoc retrieval increased both measures to 1.000 for both embedding models.

The improvement in type validity required only modest additional retrieval latency. Average latency increased from

9.2 to 11.2 ms for MiniLM and from 8.3 to 10.7 ms for BGE. Mean similarity scores remained comparable: 0.5144 versus 0.5044 for MiniLM and 0.6674 versus 0.6654 for BGE. Thus, enforcing the ontology constraint substantially improved type correctness without substantially changing the embedding similarity of the returned results.

B. Effect of Post-Hoc Candidate Budget

Figure 1 examines whether post-hoc filtering can recover the same type validity by increasing the number of unrestricted candidates available for filtering. Across both embedding models, Valid@5 generally increases as the candidate limit grows. However, the rate of improvement varies substantially by dataset, and even a candidate limit of 2,000 does not consistently recover the validity achieved by pre-hoc retrieval.

For MiniLM, mean post-hoc Valid@5 increases from 0.7413 at a candidate limit of 25 to 0.9173 at a limit of 2,000. For BGE, it increases from 0.8107 to 0.9213. In both cases, this remains below the Valid@5 of 1.000 achieved by pre-hoc retrieval.

The dataset-level behavior further shows that the difficulty of recovering type-valid results through post-hoc filtering is domain-dependent. DBpedia approaches complete validity at the largest candidate limit, reaching 0.9800 with MiniLM and 0.9920 with BGE. ATMONTO and Brick remain substantially below 1.000: at the same limit, Valid@5 reaches 0.8800 and 0.8920 with MiniLM and 0.8960 and 0.8760 with BGE, respectively.

Increasing the post-hoc candidate budget also incurs substantially greater retrieval latency. At a limit of 2,000, average post-hoc latency reaches 249.4 ms for MiniLM and 264.6 ms for BGE, compared with 11.2 ms and 10.7 ms for pre-hoc retrieval. Thus, increasing the unrestricted candidate pool can reduce type violations, but at substantially greater retrieval cost and without consistently matching the type validity obtained when the ontology constraint is supplied directly during retrieval.

Qualitative Retrieval Analysis — LaTeX

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C. Qualitative Retrieval Analysis

To examine the type errors underlying the aggregate metrics, Table III presents representative queries from the fixed benchmark together with the highest-ranked unconstrained and pre-hoc results. The examples were selected for interpretability rather than solely for the magnitude of the unconstrained error: in each case, the requested entity type is either stated directly or is readily apparent from the natural-language query. The table includes complete failures, partial failures, and queries for which unconstrained retrieval already produces a fully type-valid result set. For compactness, the embedding models are abbreviated as MiniLM and BGE.

Several examples illustrate a recurring failure mode in which unconstrained retrieval emphasizes a strongly associated contextual entity rather than the type explicitly requested by the query. For *arrival route segment near JFK*, MiniLM

returns five airport entities, producing Valid@5 = 0.0, despite the query explicitly requesting a route segment. Pre-hoc retrieval instead returns five route-segment entities. A similar effect appears in Brick. For *chiller serving ahus*, the unconstrained MiniLM results consist entirely of air handler units, while for *reheat coil on vav in smc*, unconstrained BGE retrieval returns VAV entities rather than reheat coils. In both cases, the embedding retrieves entities closely associated with the requested object but belonging to the wrong ontology class.

Ontology constraints are also useful when unconstrained retrieval is mostly, but not completely, type-correct. For the Brick query *temperature reading for supply air on AHU01 in bldg1*, four of the five unconstrained BGE results are supply-air temperature sensors, but the highest-ranked result is the AHU itself. Consequently, Valid@5 is 0.8 even though the retrieval is largely consistent with the requested type. Applying the ontology constraint removes this isolated type error and produces five valid sensor entities. This case illustrates that type violations need not appear as complete retrieval failures; they can also occur as individual high-ranking contaminants within an otherwise appropriate result set.

The examples further demonstrate that unconstrained type correctness depends on the embedding representation and the particular query. For *arrival route segment near JFK*, MiniLM obtains Valid@5 = 0.0, whereas BGE retrieves five valid route segments without a constraint. The same model dependence appears in the DBpedia hospital queries. MiniLM returns five non-hospital geographic entities for *where should I take someone who may have appendicitis*, while BGE obtains Valid@5 = 1.0 on the same query. Conversely, for *place with patients and doctors*, MiniLM returns five type-valid hospitals, whereas BGE obtains Valid@5 = 0.8: four results are hospitals, while one returned entity, Covenant Health Park, is typed as a venue/building rather than a hospital. Pre-hoc retrieval produces Valid@5 = 1.0 in each of these cases.

These examples show that natural-language similarity can often imply the intended entity type, but does not do so consistently across queries or embedding models. Ontology-constrained retrieval therefore serves a role distinct from improving the embedding representation: the embedding determines semantic ranking among eligible entities, while the ontology constraint prevents entities of an incompatible type from appearing in the returned result set. Valid@5 measures only this type-level correctness and should not be interpreted as a guarantee that every type-valid entity is otherwise the most relevant answer to the query.

VII. DISCUSSION

VIII. CONCLUSION

REFERENCES

TABLE II
RETRIEVAL PERFORMANCE ACROSS DATASETS AND EMBEDDING MODELS. POST-HOC RESULTS USE A MAXIMUM CANDIDATE LIMIT OF 2,000.

Dataset	Method	Valid@5	AllValid@5	Score	Time (ms)
<i>all-MiniLM-L6-v2</i>					
ATMONT0	Unconstrained	0.6440	0.5800	0.4724	10.6
	Pre-hoc	1.0000	1.0000	0.4589	12.7
	Post-hoc (2000)	0.8800	0.8800	0.4149	354.5
Brick	Unconstrained	0.5960	0.4800	0.6113	8.1
	Pre-hoc	1.0000	1.0000	0.6111	10.6
	Post-hoc (2000)	0.8920	0.8800	0.5616	319.9
DBpedia	Unconstrained	0.6760	0.4800	0.4593	8.7
	Pre-hoc	1.0000	1.0000	0.4432	10.4
	Post-hoc (2000)	0.9800	0.9800	0.4317	73.7
Average	Unconstrained	0.6387	0.5133	0.5144	9.2
	Pre-hoc	1.0000	1.0000	0.5044	11.2
	Post-hoc (2000)	0.9173	0.9133	0.4694	249.4
<i>bge-large-en-v1.5</i>					
ATMONT0	Unconstrained	0.7960	0.7800	0.6483	7.5
	Pre-hoc	1.0000	1.0000	0.6515	11.6
	Post-hoc (2000)	0.8960	0.8800	0.5910	296.9
Brick	Unconstrained	0.6600	0.5200	0.7437	7.7
	Pre-hoc	1.0000	1.0000	0.7420	9.2
	Post-hoc (2000)	0.8760	0.8600	0.6726	427.5
DBpedia	Unconstrained	0.6720	0.4400	0.6103	9.5
	Pre-hoc	1.0000	1.0000	0.6026	11.5
	Post-hoc (2000)	0.9920	0.9800	0.6011	69.5
Average	Unconstrained	0.7093	0.5800	0.6674	8.3
	Pre-hoc	1.0000	1.0000	0.6654	10.7
	Post-hoc (2000)	0.9213	0.9067	0.6216	264.6

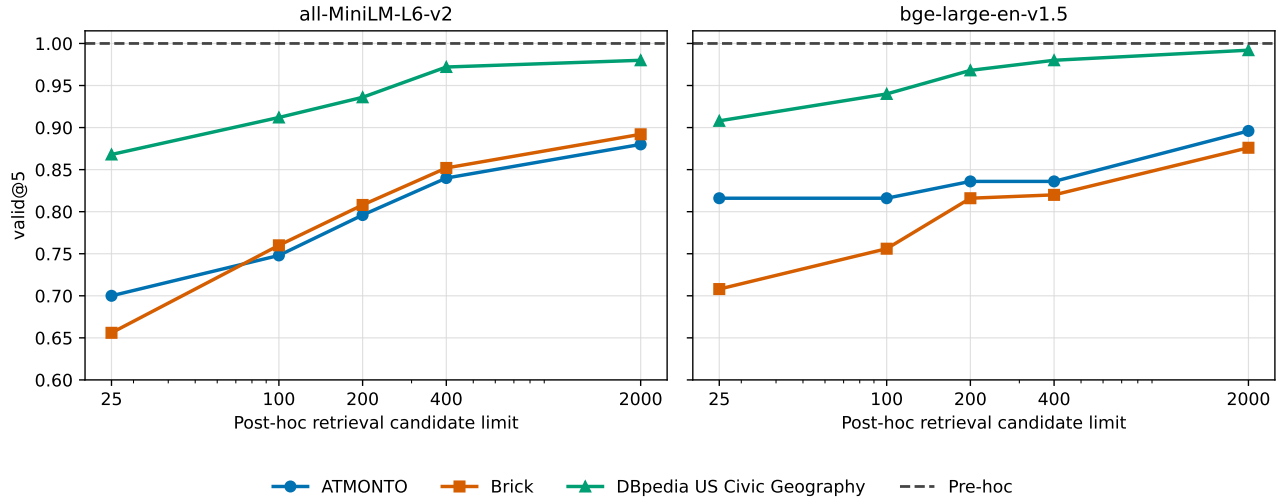


Fig. 1. Type validity under post-hoc filtering as the unrestricted retrieval candidate limit increases. Results are shown separately for all-MiniLM-L6-v2 and bge-large-en-v1.5. The dashed line denotes pre-hoc ontology-constrained retrieval, which achieves Valid@5 = 1.00 for all datasets.

TABLE III
REPRESENTATIVE QUALITATIVE RETRIEVAL RESULTS. THE FINAL COLUMN REPORTS VALID@5 FOR UNCONSTRAINED AND PRE-HOC RETRIEVAL, RESPECTIVELY.

Dataset	Model	Query	Constraint	Unconstrained top result	Pre-hoc top result	Valid@5
ATMONT0	MiniLM	<i>arrival route segment near JFK</i>	Route segment	EFKJ Airport (International airport)	J518 Subpath07620120908 01 (Route segment)	0.0 → 1.0
ATMONT0	BGE	<i>arrival route segment near JFK</i>	Route segment	rtePath20140701053_01_J53 (Route segment)	rtePath20140701053_01_J53 (Route segment)	1.0 → 1.0
Brick	MiniLM	<i>chiller serving ahus</i>	Chiller	AHU03S (Air handler unit)	chiller (Chiller)	0.0 → 1.0
Brick	BGE	<i>reheat coil on vav in smc</i>	Reheat coil	VAVRM4450 (VAV)	ReheatCoil_2-4-17A (Reheat coil)	0.0 → 1.0
Brick	BGE	<i>temperature reading for supply air on AHU01 in bldg1</i>	Supply-air temperature sensor	AHU01 (Air handler unit)	bldg1.ZONE.AHU01. RM107A.Zone Supply Air Temp (Supply-air temperature sensor)	0.8 → 1.0
DBpedia	MiniLM	<i>where should I take someone who may have appendicitis</i>	Hospital	Fredonia (Biscoe), Arkansas (City)	Albert B. Chandler Hospital (Hospital)	0.0 → 1.0
DBpedia	BGE	<i>where should I take someone who may have appendicitis</i>	Hospital	Nemours Children’s Health (Hospital)	Lake District Hospital (Hospital)	1.0 → 1.0
DBpedia	MiniLM	<i>place with patients and doctors</i>	Hospital	Southern Ohio Medical Center (Hospital)	Southern Ohio Medical Center (Hospital)	1.0 → 1.0
DBpedia	BGE	<i>place with patients and doctors</i>	Hospital	Bond Clinic (Hospital)	Bond Clinic (Hospital)	0.8 → 1.0